pfi

Python for Fixed Income

COURSE 5C · RISK MODELS

Survival Curves and Kaplan-Meier

WEEK 1 · DAY 4

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Today: when, not only whether

The cohort's development years

3,504
firms entered

345
seen to default

The rest left the study or were still going when it ended: censored.

Today's question

How do you draw a survival curve when most firms were never seen to finish?

Kaplan and Meier, 1958: use each firm for exactly the years it was watched.

Survival in study years since entry, for invented firms. Development years 1 to 14 only.

Names for a survival curve

A curve is survival in study years since entry; a hazard is the share at risk that defaulted.

Allowed names

fitted hazard

a synthetic firm's default chance

survival in study years

Never called

a rating view

a real issuer's default chance

a forecast of defaults

The credit triangle today runs on invented spreads: it says nothing about any bond.

How survival and censoring work

Survival	S(t): the share of firms still free of default t years after entry
Hazard	The share of firms at risk at the start of a year that default during it
The tie	Survival past t is the product of (1 - hazard) over every year up to t
Censored	A watch that ended without a default: left the study, or the study ended
What it tells	A censored firm survived every year it was watched, and nothing after

How the watch ended

How the watch ended	Firms	Counted as	Share
defaulted	345	an event	9.8%
left the study (refinanced or called away)	1,231	censored	35.1%
still at risk at the end of year 14	1,928	censored	55.0%

Most firms are censored. A curve that cannot use them throws away most of the data.

How Kaplan-Meier works

Distinct times	Walk through the times at which anything happened, in order
At risk	The firms watched for at least that long
The share	1 - defaults / at risk: the share that got through that time
Censored	At risk up to its last year, then out of the count; never a default
What comes out	The running product of the shares: a step curve (kaplan_meier)

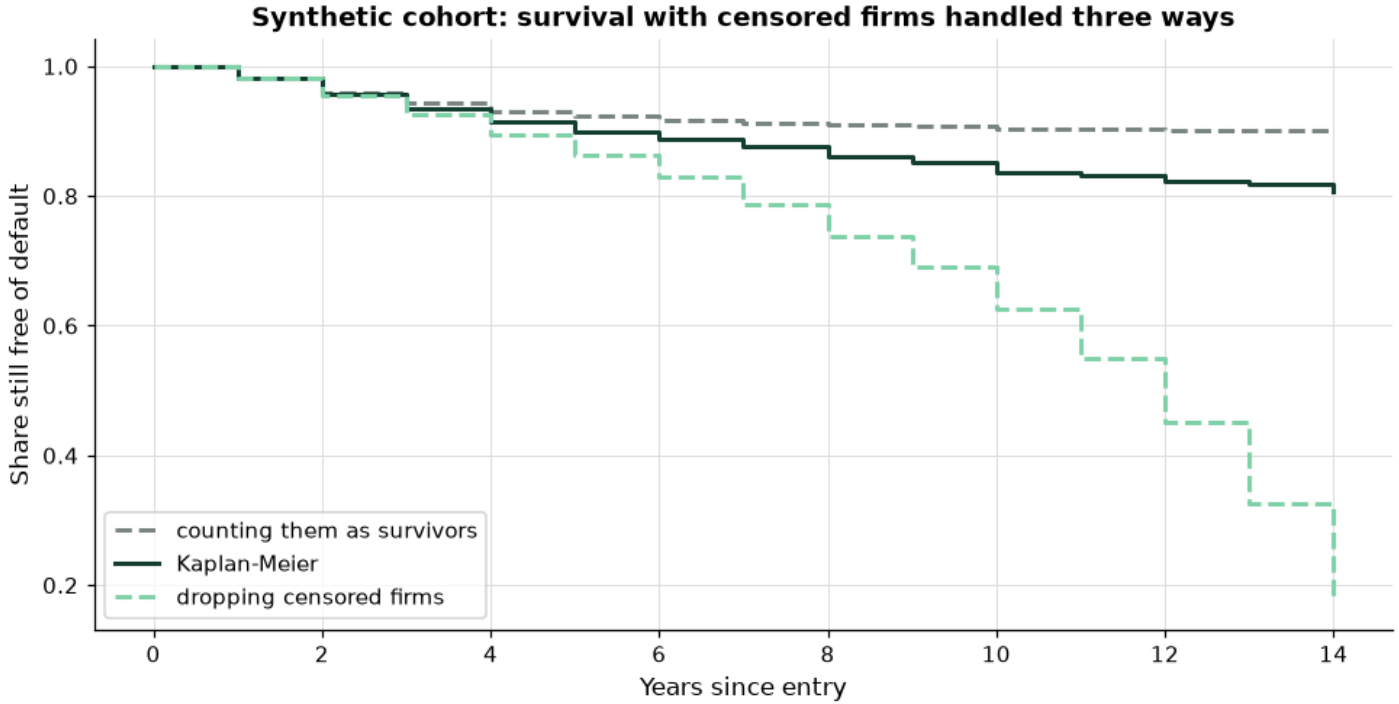
Kaplan-Meier on ten firms

Years	At risk	Defaults	Censored	Share	Survival
1	10	0	4	1.0000	1.0000
3	6	2	0	0.6667	0.6667
5	4	1	1	0.7500	0.5000
7	2	0	1	1.0000	0.5000
12	1	0	1	1.0000	0.5000

Three defaults in ten firms would say 0.7. The firms censored in year 1 cannot vouch for the later years.

A censored firm makes every later default count for more.

Why censoring matters



83.6%

Kaplan-Meier, past 10 years

62.5%

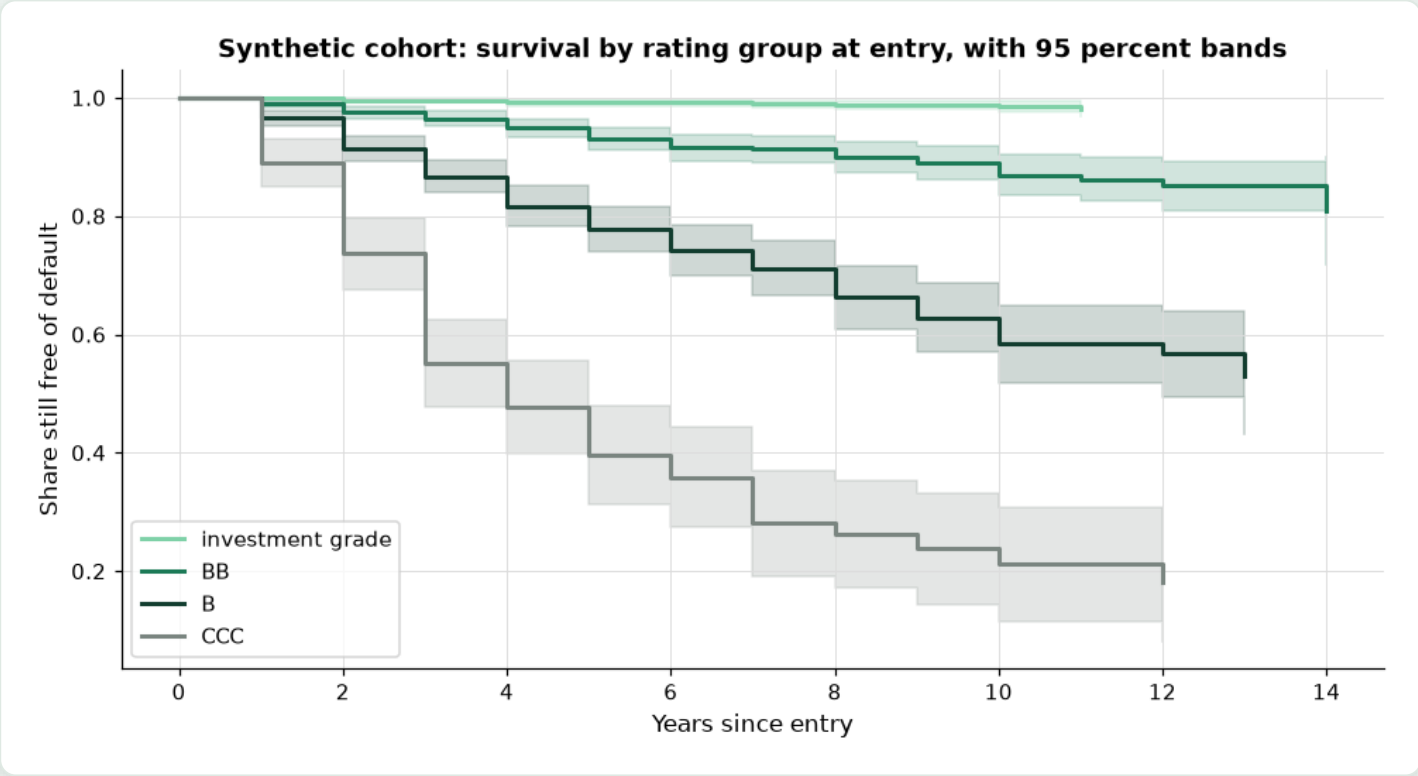
dropping them

90.3%

as survivors

Dropping removes survivors only: too low. Counting them as survivors keeps firms nobody watched: too high.

The cohort's curve, and curves by rating



Group at entry	Defaults	Past 10 yrs
investment grade	15	98.5%
BB	69	86.9%
B	149	58.4%
CCC	112	21.2%

Greenwood's 95 percent band widens as fewer firms remain at risk. These curves use the rating a firm entered with.

How the log-rank test works

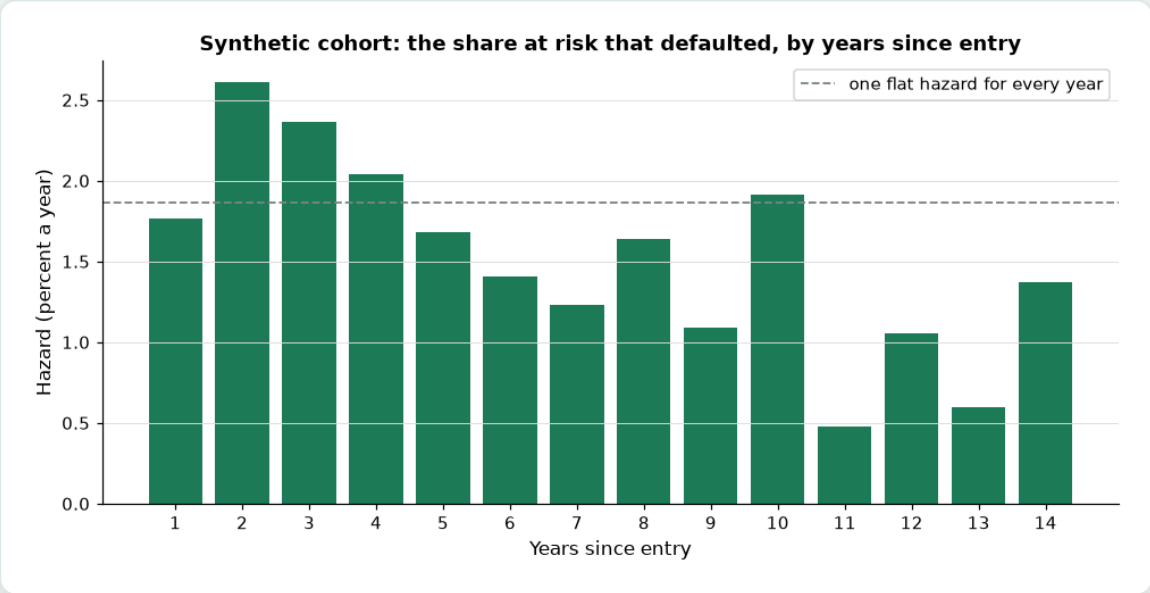
The question	Do two or more survival curves differ by more than chance would make them?
Compare	At each time with a default: the defaults each group had, against one shared curve
Add up	The gaps, given who was at risk, into a chi-squared statistic
Caveat	Every year weighs alike: curves that cross can hide from it
What comes out	A chi-squared and a p-value (statsmodels survdiff)

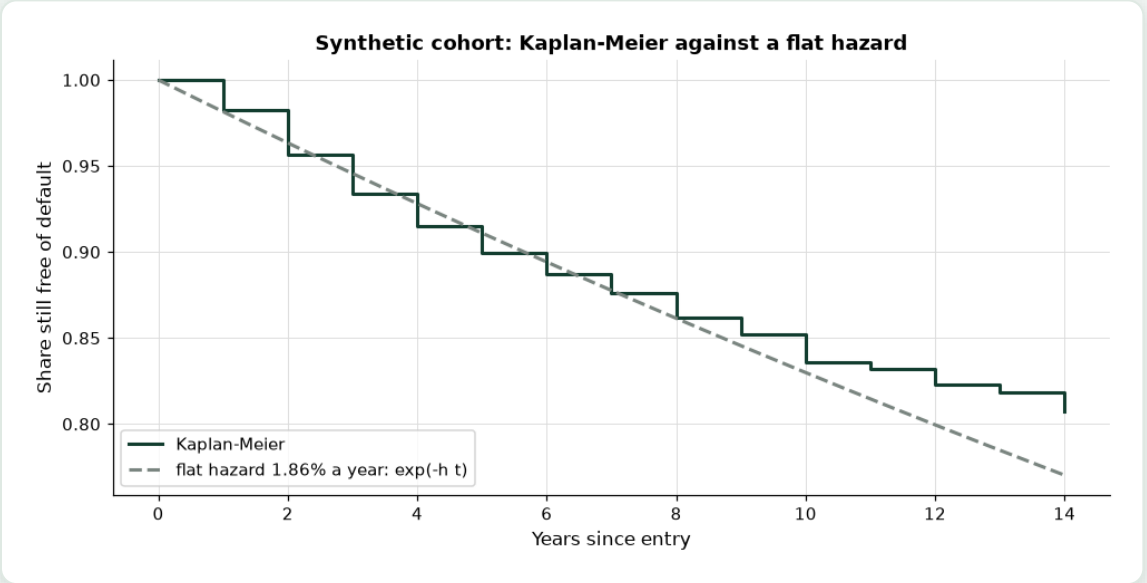
The log-rank test on the cohort

Comparison	Firms	Defaults	Chi-squared	p-value
all four groups	3,504	345	960.82	0.0000
BB against B	1,686	218	90.70	0.0000
A against BBB	1,353	15	3.67	0.0555
AA against A	680	2	0.86	0.3538

Read the defaults beside every p-value. A high p-value on two defaults proves nothing.

The hazard, and one flat hazard





One flat hazard, 1.8646% a year, is a fair summary of 14 years and a poor description of any one of them.

The credit triangle

```
spread = hazard x (1 - recovery)     so     hazard = spread / (1 - recovery)
```

Invented spread (bp)	Hazard (% a year)	Survival past 5 yrs	Default within 5 yrs
100	1.67	0.9200	8.0%
250	4.17	0.8119	18.8%
500	8.33	0.6592	34.1%

Recovery 40 percent. Arithmetic on invented numbers: a spread also pays for bearing the risk.

Two manuscripts, one paper

E. L. Kaplan and P. Meier, June 1958

"Nonparametric Estimation from Incomplete Observations"

Journal of the American Statistical Association, volume 53, pages 457 to 481

Meier began from Greenwood's paper on the duration of cancer; Kaplan from "the lifetimes of vacuum tubes in the repeaters in telephone cables buried in the ocean." The journal recommended a joint paper: "much correspondence over four years".

Source: L. J. A. Stalpers and E. L. Kaplan, BSHM Bulletin 33 (2), 2018, pages 109 to 111, quoting Kaplan in E. Garfield, Current Contents, 1983.

Excel side by side

Excel side by side (tested in Excel)

`=COUNTIF($A$2:$A$11,">="&D2)`	firms at risk: equal on every row
`=COUNTIFS($A$2:$A$11,D2,$B$2:$B$11,1)`	defaults at that time
`=1-F2/E2`	the first row's share
`=G2*(1-F3/E3)`	the running product, filled down
`=SUM(L2:L3505)/SUM(K2:K3505)`	flat hazard 1.865%
`=0.025/(1-0.4)`	triangle hazard 4.17%

Excel has no Kaplan-Meier function, no Greenwood band, and no log-rank test: the table is the way.

Save the day: read a commit back

1. Commit

mlkit.py, riskkit.py, and its tests

2. git show

Print riskkit.py exactly as the commit holds it

3. Compare

The committed file against the file in the folder

```
git show HEAD:riskkit.py
```

The committed file is the file in the folder: True.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

What is drawn, from which inputs, in which unit: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: an exit counted as a default: 1576 events from 345 defaults.

Code that runs is not the same as code that is right.

What you can do now

- Turn firm-years into one row per firm; name the censored ones
- Work Kaplan-Meier by hand and with kaplan_meier
- Show the bias of dropping censored firms
- Read curves by rating, and a log-rank test with its defaults
- Read the hazard, a flat hazard, and the triangle; 21 tests pass

Week 1, Day 5 Case study: hazard models